

DAY 02

LECTURE · 90 MINUTES

How the Internet Actually Travels.

SUBTITLE

The journey of a single click

TODAY

Packets · DNS · Cables

THEN

Routing · The edge

RECAP

Yesterday: the data centre

09:14 THIS MORNING

You tap a link. Before you can blink, something has travelled **halfway around the planet** and come back with an answer.

YESTERDAY WE STOOD INSIDE THE BUILDING. TODAY WE FOLLOW THE
CLICK THAT LEAVES IT — MILLISECOND BY MILLISECOND, ALL
LECTURE.

IT IS NOT IN THE AIR

The internet is mostly **light, racing through glass** laid along the floor of the ocean.

Five questions, **one click.**

TODAY'S LECTURE

§ 01

What actually leaves your device?

Packets, addresses, and the postal system hiding inside every connection.

§ 02

How does a name become a number?

DNS, the quiet phone book the whole internet depends on.

§ 03

Where does it physically go?

Undersea cables, the speed of light, and why distance still rules.

§ 04

Who decides the route?

A network of networks, held together by announcements and trust.

§ 05

How is it ever this fast?

CDNs, and the trick of keeping a copy close to everyone.

— SECTION ONE

01

The journey of **a click.**

Nothing about the internet is magic. It is **a very fast, very clever postal service.**

• THE CLICK • T+0 MS • IT HAS JUST BECOME A BURST OF PACKETS

THE ONE IDEA THAT RUNS EVERYTHING

Everything you send is chopped into tiny envelopes called **packets**. A page, a song, a video call: all of it, packets.

GET THIS ONE PICTURE AND THE REST OF THE INTERNET STOPS BEING MYSTERIOUS.

So what is in **one packet**?

LIKE ANY ENVELOPE, TWO PARTS

THE HEADER

The address label

Who it is from, who it is going to, and which piece of the message this is.
Routers read only this.

THE PAYLOAD

The contents

A small slice of the actual thing you are sending. One packet might hold a sliver of an image, or a fraction of a second of audio.

A single web page can break into **thousands** of these, each one a self-contained little envelope finding its own way.

Why cut it up **at all?**

THREE REASONS, ALL ABOUT RESILIENCE

-
- 01 **Each piece can take its own road**
If one route is busy or broken, individual packets simply detour around the trouble. No single jam stops the whole message.
-
- 02 **Lose one, resend just one**
If a packet goes missing, only that tiny envelope is sent again, not the entire file. Cheap to recover.
-
- 03 **Everyone shares the same pipes**
Your packets and a million other people's are interleaved down the same cables, taking turns microsecond by microsecond.
-

EVERY DEVICE HAS A NUMBER

A phone number **for machines.**

For one device to send a packet to another, it needs an address. That address is the **IP address**, a string of numbers every connected device carries.

We ran out of the original supply years ago. The newer scheme, **IPv6**, has room for **more addresses than there are grains of sand on Earth.**

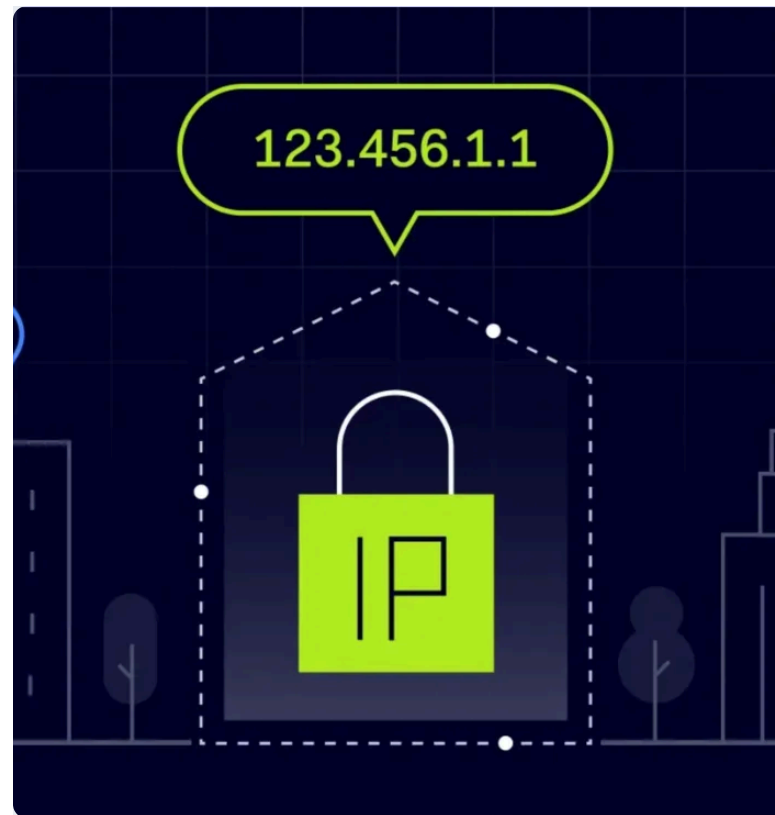


FIG. WHERE THE ADDRESSES LIVE

WRAPPED IN LAYERS

L4 Application

L3 Transport · TCP

L2 Internet · IP

L1 Link · Wi-Fi / fibre

LAYER 4 / APPLICATION



The part you actually mean: a plain request like “send me the cat photo.” Human-readable, and useless until it is wrapped for travel.

A SMALL MIRACLE AT THE OTHER END

Packets arrive out of order, by different roads, at different times. Your device quietly **sorts them back into one whole.**

THE NETWORK ITSELF REMEMBERS NOTHING. IT JUST FORWARDS ENVELOPES. ALL THE BOOKKEEPING HAPPENS AT THE TWO ENDS.

Don't believe it? **Send something.**

SEND THE PAGE

ONE PAGE, TWELVE PACKETS, THREE ROADS

ROUTE A · QUIET

ROUTE B · CONGESTED

ROUTE C · THE LONG WAY

REASSEMBLY BUFFER · YOUR DEVICE

01

02

03

04

05

06

07

08

09

10

11

12

THE POLITE PROTOCOL

TCP says “**hello**” first.

Before sending anything important, two devices exchange a quick three-step greeting: “**Can you hear me? Yes, can you hear me? Yes.**” Only then does the data flow.

And every packet that arrives gets a little nod of confirmation. No nod, and the sender **tries again**. That is how the internet is reliable on top of pipes that are not.

TCP/IP

3 Way Handshake

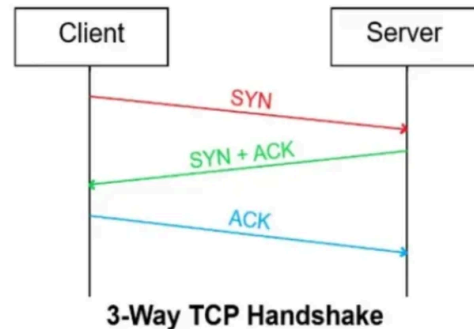


FIG. THE THREE-WAY HANDSHAKE

Sometimes losing one is **fine**.

TWO DELIVERIES, TWO PHILOSOPHIES

TCP

The **courier**.

Signs for every envelope, resends anything missing, hands it all over in order. Pages, files, messages: **anything that must arrive whole.**

UDP

The **firehose**.

Just keeps sending. No receipts, no resends, no waiting. A lost syllable in a call is not worth going back for: **fresh matters more than complete.** Live video, calls, games.

• STORY • 1969

The first message ever sent was “LO.”

In 1969, a student at UCLA tried to log in to a computer at Stanford over the brand-new ARPANET, the internet’s ancestor. He started typing **L, O, G...**

The system crashed after the second letter. So the very first thing the internet ever said was, fittingly, “**LO.**” As in, lo and behold.

The lesson: everything you do online is built on a network that, on day one, could barely say hello.



Interface Message Processor (IMP) 1969

— SECTION TWO

02

Names & numbers.

You type words. The network only understands numbers. **Something has to translate.**

- THE CLICK • T+1 MS • IT KNOWS A NAME, BUT NOT YET A NUMBER

A VERY HUMAN PROBLEM

Nobody remembers a friend by their phone number. You remember a **name**. The internet has exactly the same problem.

YOU THINK IN NAMES LIKE KAUST.EDU.SA. MACHINES NEED A NUMERIC ADDRESS. BRIDGING THE TWO IS A JOB DONE BILLIONS OF TIMES A SECOND.

Read a web address **right to left.**

IT IS A POSTAL ADDRESS, NARROWING DOWN

.sa

The top-level domain

The broadest bucket. The country or category, like .sa, .com or .org. Read it first, like a country on an envelope.

edu

The kind of organisation

A subdivision within .sa, here marking an educational institution.

kaust

The organisation itself

The specific name that an owner registered. This is the part people actually remember.

www

The machine or service

Which server at KAUST you want. The web, mail, or any other service living under that name.

THE INTERNET'S DIRECTORY

DNS is the **phone book**.

The Domain Name System is one enormous, shared directory. Give it a name, and it hands back the number.

No single computer holds the whole book. It is **split across the world** and stitched together so that, to you, it always feels like **one instant lookup**.

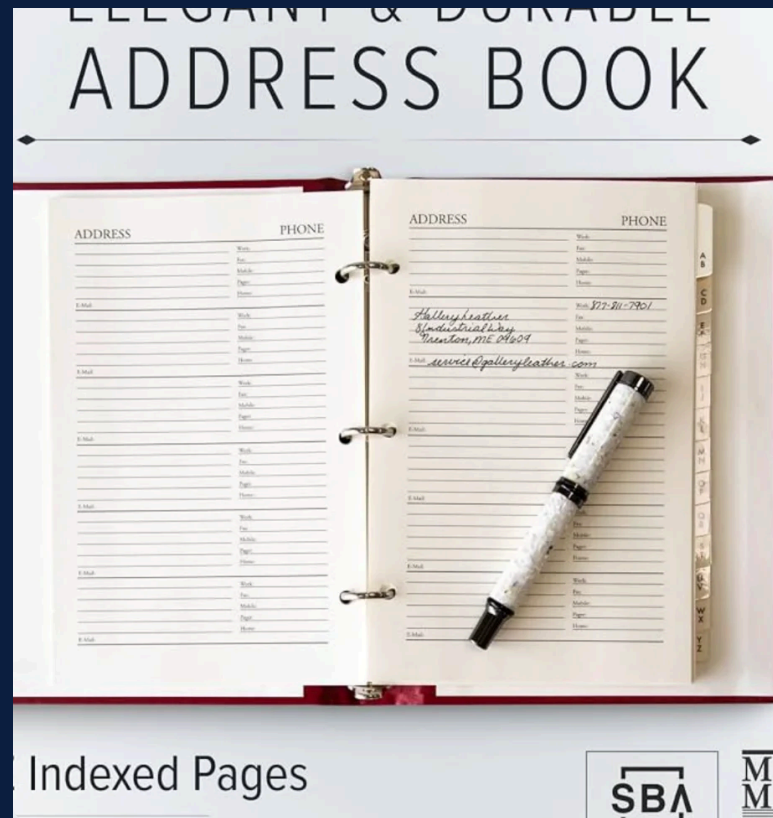


FIG. ONE BOOK, EVERYWHERE AT ONCE

Watch a name become a number.

ALL OF THIS, IN A FEW THOUSANDTHS OF A SECOND

01

You

“Where is kaust.edu.sa?”

02

Resolver

Your provider's helper does the asking around for you.

03

Root

“Not sure, but go ask whoever runs .sa.”

04

.sa server

“Ask the name server for kaust.edu.sa.”

05

Authoritative

“Here is the real address.”

▷ Look it up again

WHY THE SECOND VISIT IS INSTANT

Once an answer comes back, it is **remembered** for a while. Your device, your provider, everyone in between keeps a copy.

THIS IS CACHING, AND IT IS EVERYWHERE IN COMPUTING: DO THE SLOW WORK ONCE, THEN REUSE THE ANSWER. WE WILL SEE THE VERY SAME TRICK POWER CDNS AT THE END OF TODAY.

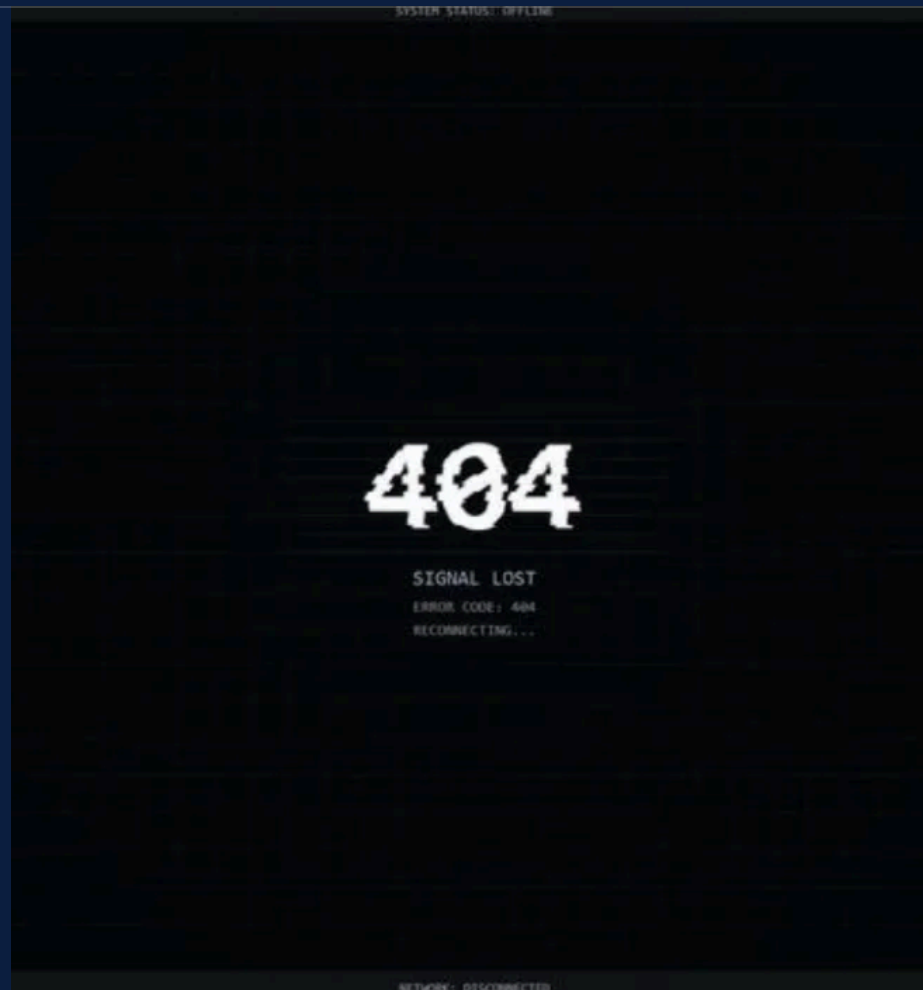
• STORY • OCTOBER 2021

The day a giant **erased itself** from the phone book.

On October 4, 2021, a routine command inside Facebook's network disconnected all its data centres from each other — and its own DNS servers withdrew Facebook, Instagram and WhatsApp from **the internet's directory**.

For six hours, roughly **3.5 billion people** could not connect. Staff could not even badge into the buildings — the badge readers ran on the same network that had just vanished.

The lesson: when your recovery plan depends on the network you're recovering, you don't have a backup plan. You have one plan.



— SECTION THREE

03

The physical **internet**.

Strip away the software and you are left with something startlingly real: **glass, ships, and the ocean floor.**

• THE CLICK • T+45 MS • SOMEWHERE UNDER THE RED SEA, TRAVELLING AS LIGHT

WHERE INTERCONTINENTAL TRAFFIC REALLY GOES

99%

Almost all data crossing an ocean travels not by satellite, but through **cables on the seabed**. Hundreds of them, criss-crossing the planet, carrying nearly everything between continents.

So why not just use **satellites?**

FOR THE HEAVY LIFTING, GLASS STILL WINS

CABLES

Fat, cheap, **close.**

A single cable carries a staggering amount of data, and the light inside has only the ocean floor to cross. **Lower delay, far more capacity.**

SATELLITES

Far, and **getting closer.**

Traditional satellites sit so high that the round trip adds real delay. New low-orbit fleets help in remote places, but cables still move the bulk of the world's traffic.

THE SIZE OF THE THING CARRYING EVERYTHING

a garden hose

That is roughly how thick a transoceanic cable is. The glass fibres inside, the part actually carrying the light, are each **thinner than a human hair**. The rest is just armour against anchors and sharks.

FROM JEDDAH TO...

Riyadh	850 km
London	4,800 km
New York	10,400 km
Tokyo	9,000 km
Sydney	11,500 km

104_{ms}

ROUND TRIP • THE PHYSICAL FLOOR

A signal to **New York** and back covers 20,800 km. Even at two-thirds the speed of light in glass, that is the **floor**. Real networks, with their detours and queues, only ever add to it.

WHERE NETWORKS MEET

The internet's **handshakes.**

Cables have to connect to each other somewhere. That somewhere is an **Internet Exchange Point**, a building where hundreds of networks plug into a common fabric.

Hand traffic over locally and it never has to make a long, expensive detour. A handful of these buildings quietly carry **a huge share of the world's traffic.**

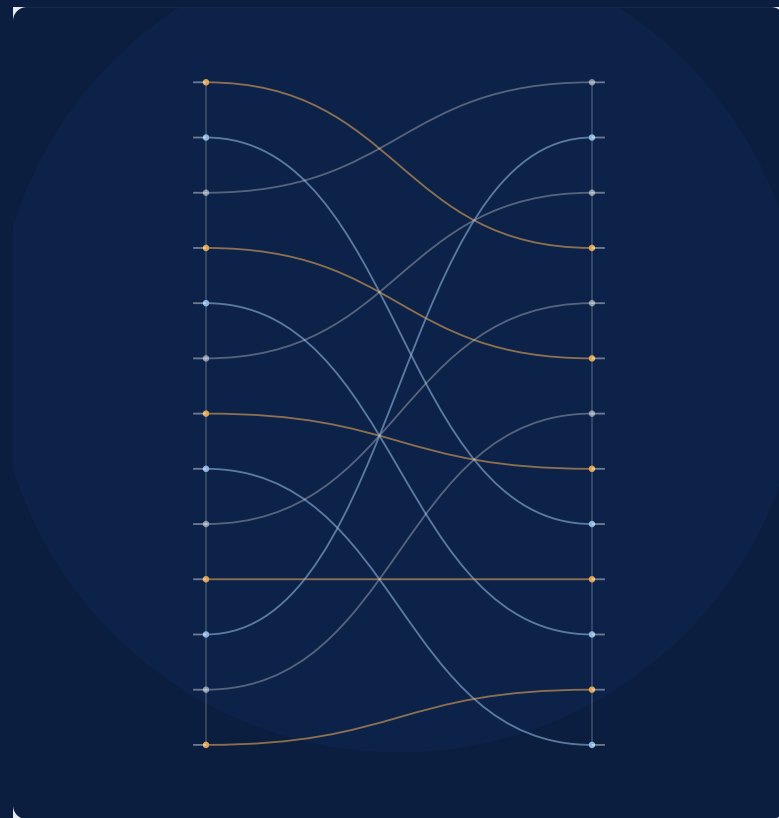


FIG. THE MEET-ME ROOM

Now break it. **Cut a cable.**

JEDDAH TO SINGAPORE, THREE WAYS ACROSS

CUT THE DIRECT CABLE

88_{ms}

ROUTE A · DIRECT SUBSEA

CARRYING TRAFFIC

ROUTE B · VIA MUMBAI

STANDBY

ROUTE C · THE LONG WAY

STANDBY

● ALL PACKETS ARRIVING · JEDDAH ⇄ SINGAPORE

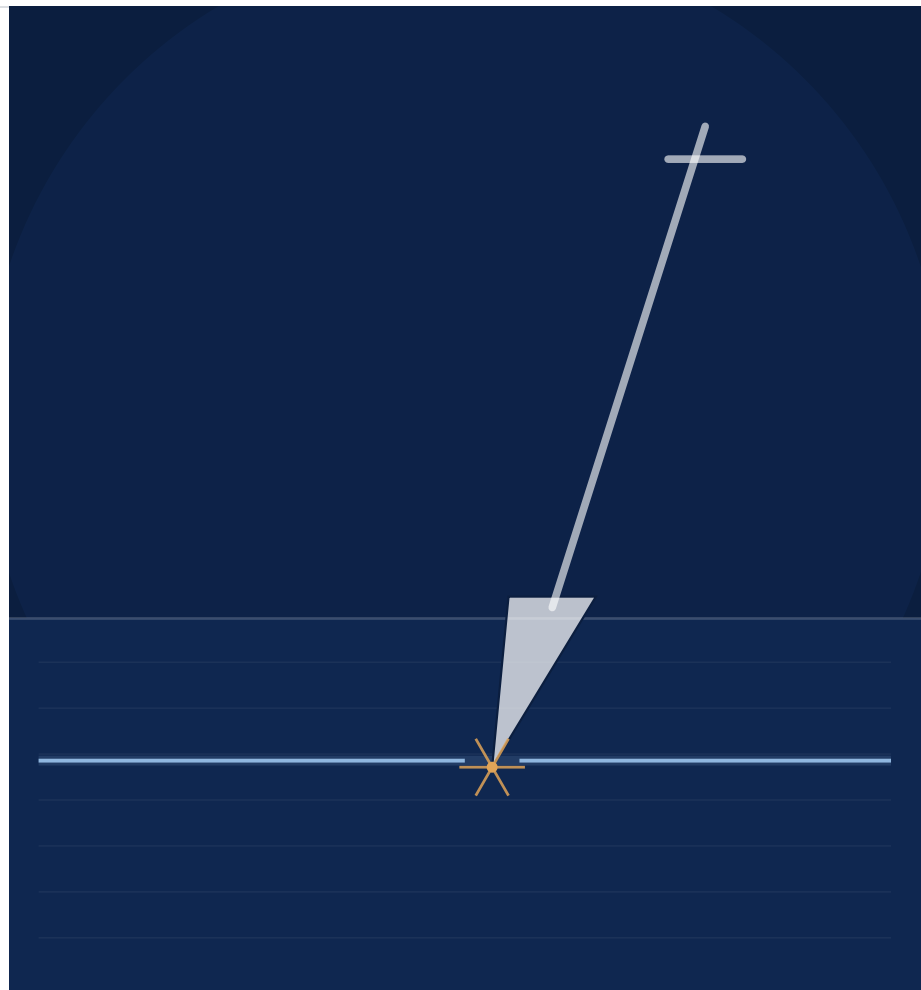
• STORY • 2011

One woman with a spade cut off a whole country.

Scavenging for copper to sell, a pensioner in Georgia drove her spade into the ground and sliced through a fibre cable. In an instant, most of **neighbouring Armenia** lost the internet.

An entire nation, offline for hours, because a single buried strand of glass happened to run through one field.

The lesson: the internet feels boundless and abstract, but it rests on physical things that a spade, an anchor, or a storm can sever.



— SECTION FOUR

04

The politics of **a packet.**

No one runs the internet. So how does any packet ever know **which way to go?**

• THE CLICK • T+70 MS • CROSSING ITS FOURTH NETWORK; NOBODY IS IN CHARGE

A SURPRISING FACT

There is no master computer running the internet. It is **tens of thousands of separate networks** that simply agree to connect.

YOUR HOME PROVIDER, A UNIVERSITY, A BANK, A HYPERSCALER: EACH RUNS ITS OWN NETWORK, AND THE INTERNET IS JUST WHAT HAPPENS WHEN THEY ALL LINK UP.

THE BUILDING BLOCK OF THE INTERNET

Each network is an **island**.

Every big network, called an **autonomous system**, gets its own number and runs itself however it likes inside its own borders.

The internet is these islands building bridges to one another, then constantly telling each other **which destinations they can reach**.

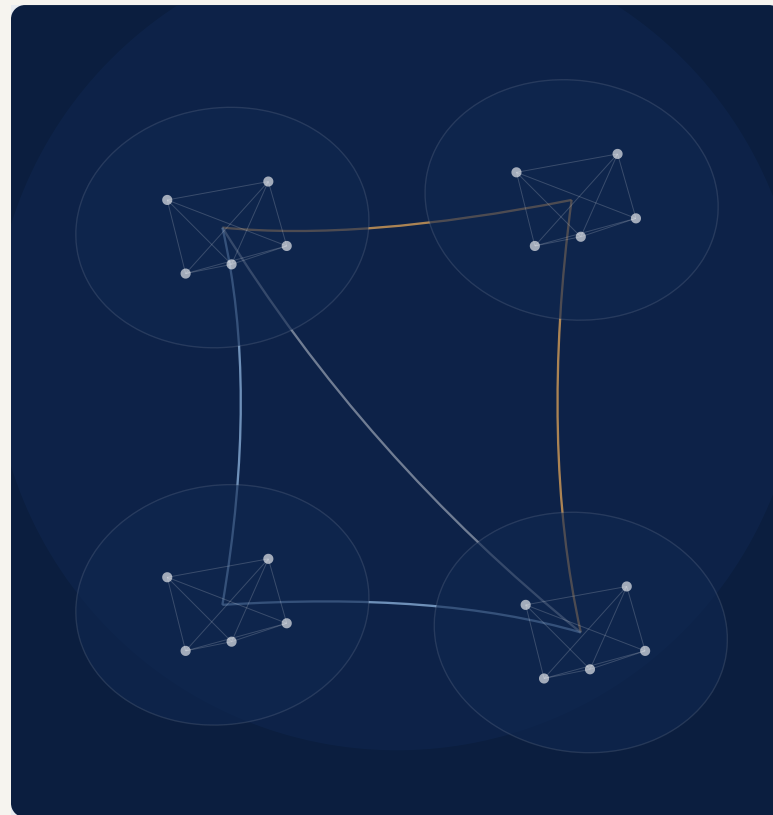


FIG. ISLANDS AND BRIDGES

THE PROTOCOL THAT GLUES IT TOGETHER

Networks shout to their neighbours, **“I can reach these addresses, send them my way.”** That endless conversation is called BGP.

THERE IS NO REFEREE CHECKING THESE CLAIMS. THE WHOLE SYSTEM RUNS, ASTONISHINGLY, ON NETWORKS CHOOSING TO BELIEVE ONE ANOTHER.

Two ways networks **connect.**

THE QUIET ECONOMICS BEHIND EVERY ROUTE

PEERING

A handshake.

Two networks agree to swap each other's traffic directly, usually **for free**. Shorter path, faster for users, cheaper for both.

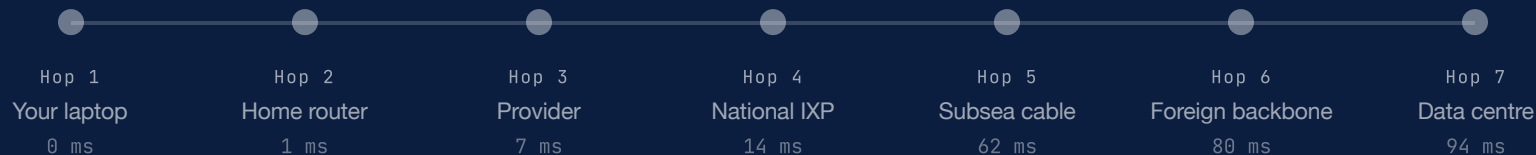
TRANSIT

A toll road.

A smaller network **pays** a bigger one to carry its traffic onward to the rest of the internet. Simple to buy, but you pay by the gigabyte.

Follow one packet, **hop** **by hop.**

THIS IS EXACTLY WHAT TODAY'S LAB WILL SHOW YOU



0ms ACROSS THE WORLD AND BACK TO YOU

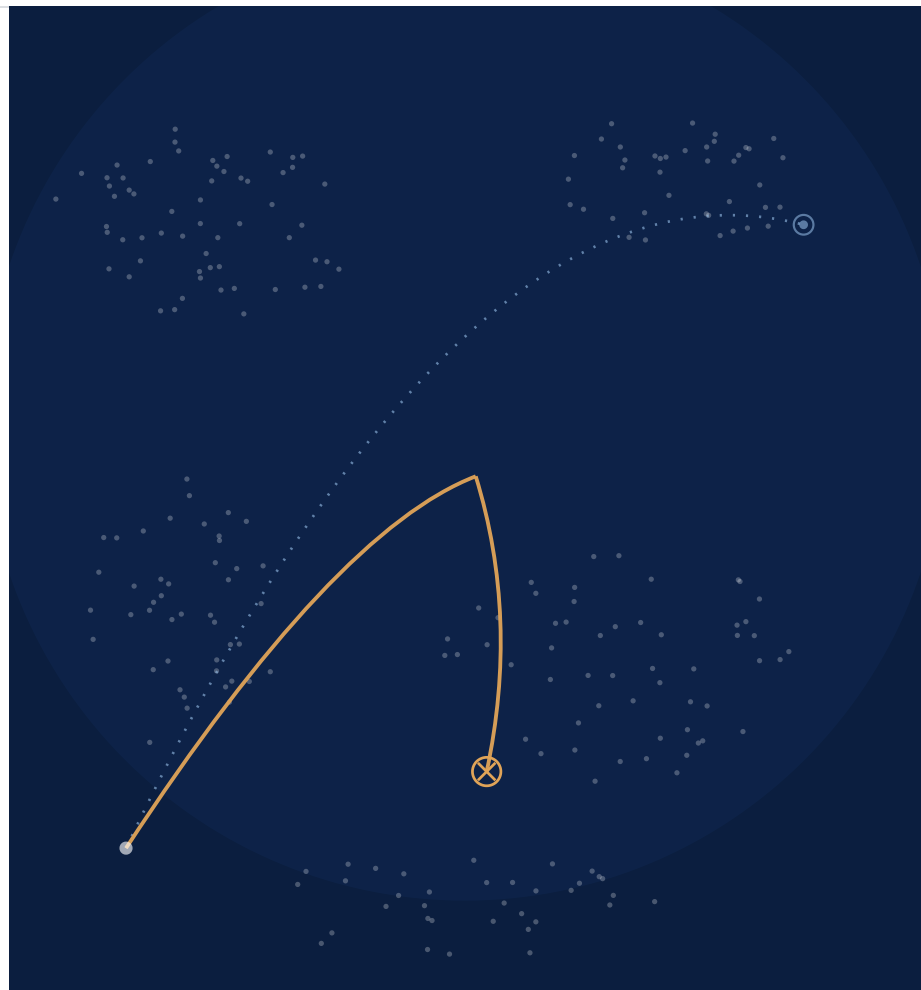
• STORY • 2008

One country tried to block a site, and broke it **for everyone.**

To censor a video at home, a telecom in Pakistan announced to the world that **it** was the right path to YouTube. Because BGP runs on trust, networks everywhere believed it.

For a couple of hours, a large slice of the planet's YouTube traffic was funnelled into a dead end in Pakistan. A local action became a **global outage**.

The lesson: the internet's openness is also its weak spot. One confident, wrong announcement can ripple worldwide.



— SECTION FIVE

05

Bringing it **closer.**

If distance is the enemy, there is one obvious move: **do not be far away.**

• THE CLICK • T+140 MS • THE ANSWER IS ALREADY ON ITS WAY BACK TO YOU

THE ONE PROBLEM YOU CANNOT ENGINEER AWAY

You cannot make light go faster. So the only way to cut the delay is to **shorten the distance.**

WHICH RAISES A STRANGE IDEA: WHAT IF A COPY OF THE WEBSITE YOU WANT WAS ALREADY SITTING IN A BUILDING JUST A FEW KILOMETRES FROM YOU?

COPIES, EVERYWHERE

Keep it **near everyone.**

A Content Delivery Network keeps copies of popular content, videos, images, whole pages, in **hundreds of cities** at once.

When you load a site, you are very often not reaching its home at all. You are pulling it from a copy **a short hop away**. That is why the internet feels instant.

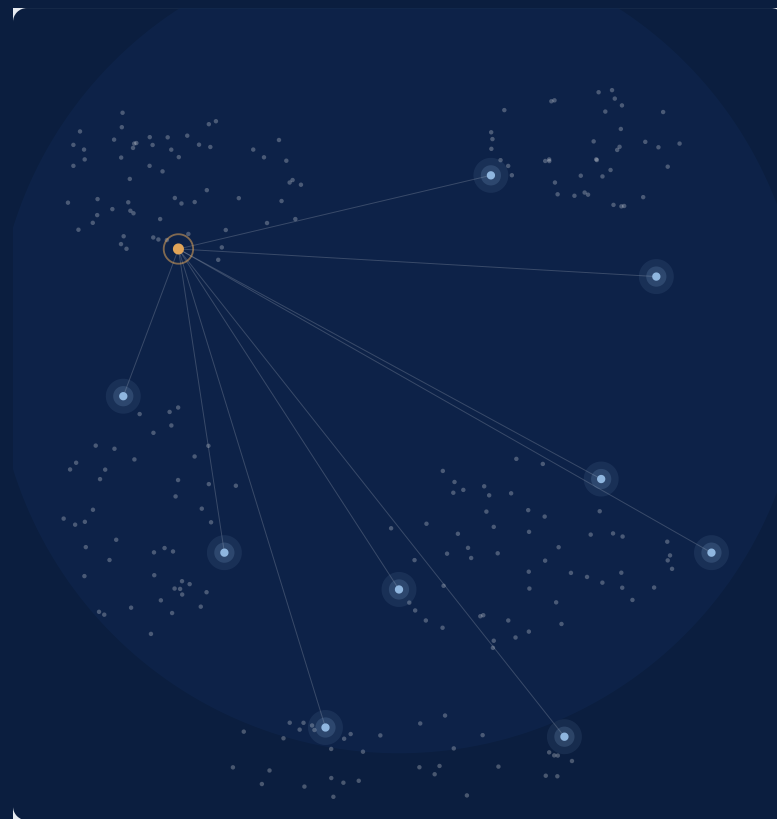


FIG. CONTENT, CLOSE TO YOU

One address, **many** places.

THE TRICK THAT PICKS THE NEAREST COPY FOR YOU

01

The same address lives in dozens of cities

Through a trick called anycast, identical copies all answer to one address, scattered around the world.

02

The network sends you to the closest one

Routing naturally carries your request to whichever copy is nearest, without you ever choosing.

03

If one location fails, the rest absorb it

Traffic simply flows to the next nearest copy. The same idea that makes it fast also makes it sturdy.

Feel the difference.

THE SAME PAGE, SERVED FROM 11,000 KM AWAY VS 40 KM AWAY

LOAD BOTH PAGES

WITHOUT A CDN • ORIGIN SERVER, VIRGINIA



0ms

WAITING

WITH A CDN • EDGE COPY, JEDDAH



0ms

WAITING

A HINT OF WHERE WE ARE HEADED

At first these copies held only images and video.
Now they are starting to run **actual code.**

WHEN COMPUTING ITSELF MOVES OUT TO THOSE THOUSANDS OF NEARBY
LOCATIONS, IT GETS A NAME OF ITS OWN: THE EDGE. THAT IS DAY
04.

— THE REFRAME

A click is a **journey**: cut into packets, named by DNS, carried through glass under the sea, routed on trust, and cached a short hop from you.

Total elapsed: about a quarter of a second. You will never see a loading bar the same way again. **Tomorrow**: what it means to live in the cloud, where this whole journey ends up running on rented machines.